

SAMS: A Mobile-Based Student Assistant Management System

A Capstone Project

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In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Information Technology with

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CHAPTER 1

INTRODUCTION

Information systems have become essential in modern education because they improve efficiency, transparency, and accessibility in school operations. Many higher education institutions adopt digital platforms to streamline processes such as enrollment, attendance, and student services. In the Philippine context, the shift toward paperless workflows is reinforced by national policy recognizing electronic documents and digital signatures as valid for official transactions when supported by proper authentication and security mechanisms [1].

One area where digital systems can provide significant benefits is the management of student employment programs, particularly student assistantships. Student assistants support administrative offices and campus activities, but handling applications, schedules, and attendance often requires substantial administrative effort when processes remain manual. QR-based attendance systems have been shown to improve efficiency and accuracy compared to traditional logging methods [2]. However, literature also emphasizes that QR-based attendance should be strengthened with authentication controls to ensure integrity and prevent unauthorized or proxy logging [3]. To address these concerns, multi-factor authentication (MFA), including OTP-based validation, is widely recommended to reduce the risk of unauthorized access by introducing layered verification [4], [5]. Time-based OTP mechanisms with limited validity periods further strengthen protection against replay and impersonation attacks [6].

At NU Lipa, the Student Assistant Program is still managed through manual processes. Applicants submit printed requirements, wait for evaluation, and once accepted, record daily time-in and time-out on paper. These practices can lead to delays, errors, and challenges in monitoring attendance, performance, and duty hour reporting. While existing attendance solutions improve logging efficiency, many systems remain focused only on attendance and do not integrate duty scheduling, document handling, and monitoring tools into one platform [2]. This gap highlights the need for an integrated system that supports end-to-end student assistant management.

To address these limitations, the researchers propose a Mobile-Based Student Assistant Management System with a Web-Based Administrative Dashboard, hereafter referred to as the Student Assistant

Management System (SAMS). The system provides a mobile application that allows students to apply for the program, view assigned duties, and record attendance digitally. The system also includes a web-based administrative dashboard designed for the Student Development and Activities Office (SDAO) and office heads features that are better optimized for larger screens, such as detailed dashboards, comprehensive schedule views, and consolidated attendance and duty-hour reports.

This platform enables administrators to monitor applications, manage duty schedules, evaluate performance, and access real-time attendance data efficiently. To further enhance operational efficiency, the system incorporates automated requirement filtering to ensure that only qualified applicants proceed in the selection process, as well as an algorithm-guided recommendation feature that assists administrators in assigning student assistants to offices based on skills, availability, and submitted schedules. Collectively, these features aim to reduce manual workload, minimize errors, and promote fairness, transparency, and consistency in duty assignments.

1.1 Project Context

The Student Assistant Program at NU Lipa currently relies on manual processes such as printed applications, paper-based attendance logs, and handwritten schedules. Administrators spend considerable time verifying application requirements, assigning duties, and monitoring student assistants' attendance. Student assistants record their daily time-in and time-out manually, which may result in errors, inconsistencies, and difficulty in validating attendance records. For administrators, consolidating and reviewing these paper-based records requires additional effort and time, often causing delays in evaluation, monitoring, and reporting.

Recent literature supports the shift toward paperless administrative workflows, especially when electronic documents are strengthened by appropriate authentication and security controls [1]. Similarly, QR-based attendance systems can improve efficiency and reliability [2], but must be supported by stronger verification measures to prevent misuse [3]. Security best practices further recommend MFA approaches such as OTP validation to reduce account compromise and unauthorized transactions [4], [5], while time-limited OTP mechanisms strengthen protection against impersonation and replay attacks [6].

In response, the proposed SAMS provides a mobile platform for students and a web-based administrative dashboard for SDAO and office heads that integrates secure attendance monitoring,

scheduling, document management, and performance evaluation. The system combines QR-based attendance logging with PIN/OTP validation to reduce proxy attendance risks [3]–[6]. It also provides an administrative dashboard that supports real-time monitoring and consolidated reporting to reduce manual administrative workload, aligned with the efficiency gains documented in QR-enabled attendance solutions [2]. In addition, the system supports digital document workflows (e.g., MOA uploads, NOA generation) consistent with the direction of recognized electronic document practices [1].

1.2 Objectives of the Study

General Objective

To design and develop a mobile-based system for student assistants and a web-based administrative dashboard that automates the application, scheduling, and monitoring of student assistants at NU Lipa.

Specific Objectives

To achieve the general objective, the study will pursue the following:

- Develop a mobile-based platform that allows students to submit applications, view duty schedules, and monitor attendance records, and access consolidated reports.
- Implement a scheduling and assignment module that validates duty assignments against students' academic schedules and supports fair, skill-based, and availability-based deployment of student assistants.
- Design and implement a secure digital attendance monitoring system using QR code scanning integrated with PIN and One-Time Password (OTP) validation to ensure accurate and reliable time-in and time-out recording.
- Develop a web-based administrative dashboard for SDAO and office heads to efficiently manage applications, verify requirements, monitor attendance and duty hours, and evaluate student assistant performance.

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- ~~Integrate digital document management and reporting features, including online~~

submission of requirements, automated generation of MOA and NOA with PDF outputs, and printable duty-hour and performance reports.

- To evaluate the system using the ISO/IEC 25010 Product Quality Model, focusing on selected quality characteristics such as functional suitability, usability, performance efficiency, reliability, and security.

1.3 Scope and Delimitation

Scope of the Project

The study covers the design and development of a mobile-based system for student assistants and a web-based administrative dashboard for NU Lipa's Student Assistant Program under the supervision of SDAO and office heads. The system includes digital application and requirement submission, duty scheduling and assignment, secure attendance monitoring using QR scanning with PIN/OTP validation, real-time monitoring through a web dashboard, document management (MOA upload, NOA generation, PDF outputs), and performance evaluation tools. These inclusions are designed to remain feasible within one academic term and within available development resources.

Delimitation of the Project

The project is limited to operational management of the Student Assistant Program under NU Lipa's SDAO. Payroll processing and financial disbursement are excluded because these functions are handled by the university's accounting system and require sensitive financial integration beyond the scope of this study. Integration with other academic platforms such as grading, enrollment, or institutional scheduling systems is also excluded. Evaluation of the system is limited to usability and efficiency testing within one academic term due to time and resource constraints.

CHAPTER 2

REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

2.1 RELATED LITERATURES/STUDIES

Digital Transformation in Higher Education and Paperless Administrative Workflows

The increasing reliance of higher education institutions on information systems reflects a global shift toward operational efficiency, transparency, and improved accessibility of academic and administrative services. Universities adopt digital platforms to streamline processes such as enrollment, monitoring, and student support services, primarily to reduce administrative workload and improve data accuracy and consistency. In the Philippine context, this transformation is reinforced by national policies recognizing electronic documents and digital signatures as valid for official transactions, provided that authentication and security mechanisms are properly implemented [1].

This policy environment strengthens the justification for digitizing administrative processes that were traditionally paper-based, particularly for student support programs that require document submission, verification, and record retention. Paperless workflows enable faster application processing, reduce the risk of document loss, and provide traceable and auditable records [2]. In line with these developments, the proposed Student Assistant Management System (SAMS) supports digital document workflows such as Memorandum of Agreement (MOA) uploads, Notice of Acceptance (NOA) generation, and standardized PDF outputs, consistent with recognized electronic documentation practices [1], [3].

2.2 Operational Challenges in Manual Student Assistant Program Management

Student assistantship programs play a vital role in supporting administrative offices and campus operations. However, when these programs rely on manual processes, they introduce significant administrative challenges. At NU Lipa, the student assistant program remains largely paper-based: applicants submit printed requirements, undergo manual evaluation, and record daily time-in and time-out entries using attendance logbooks. Such practices are prone to delays, human error, and inconsistencies, making it difficult to monitor attendance, performance, and accumulated duty hours accurately.

Manual consolidation of attendance and duty records further increases administrative burden, particularly when monitoring is conducted across multiple offices and numerous student assistants. Studies indicate that fragmented and paper-based administrative systems limit visibility and slow down decision-making processes [2], [4]. This operational context highlights the need for an integrated platform that supports end-to-end student assistant management—from application intake and requirement verification to duty scheduling, attendance monitoring, and report generation. The proposed SAMS directly addresses these challenges by integrating these functions into a unified web- and mobile-based system accessible to both students and administrators..

2.3 QR-Based Attendance Systems: Efficiency Benefits and Integrity Risks

Attendance monitoring is a critical component of student assistant programs because it directly affects accountability and duty-hour computation. Several studies indicate that QR-based attendance systems improve efficiency and accuracy compared to traditional manual logging methods. These systems minimize encoding errors, accelerate attendance recording, and support structured data storage suitable for automated reporting and monitoring [5], [6].

Despite these advantages, literature also highlights integrity risks associated with QR-based attendance systems when authentication mechanisms are insufficient. Identified risks include unauthorized scanning, proxy attendance, and QR reuse through screenshots or shared images [7]. As a result, researchers emphasize that QR-based attendance mechanisms should be strengthened with additional verification layers to ensure the validity and trustworthiness of attendance records [5], [7].

In response to these findings, SAMS adopts QR-based attendance logging to enhance efficiency while explicitly incorporating layered verification mechanisms to protect data integrity, particularly because attendance records serve as inputs for evaluation, monitoring, and administrative decision-making.

2.3.1 Multi-Factor Authentication (MFA) and OTP Validation for Secure Transactions

Systems that handle attendance data, scheduling assignments, and identity-linked records are especially vulnerable to unauthorized access and impersonation. Multi-factor authentication (MFA) is widely recommended as a best practice for reducing account compromise by requiring multiple layers of verification beyond a single credential [8]. One-Time Password (OTP) mechanisms are commonly cited as effective MFA implementations for strengthening system

security [9].

Time-based OTPs provide additional protection by limiting credential validity, thereby reducing the risk of replay and impersonation attacks. In attendance monitoring contexts, OTP expiration is particularly important because attendance transactions are time-sensitive and should represent real-time user presence rather than reusable credentials [10]. In SAMS, OTPs are configured with short expiration windows and validated against active sessions, addressing risks such as phone borrowing, impersonation, and proxy attendance.

Overall, the security architecture of SAMS aligns with reviewed recommendations: QR codes enhance logging efficiency, while PIN and OTP validation strengthen user identity assurance during attendance recording [7]–[10].

2.3.2 Algorithm-Guided Scheduling and Assignment: Fairness, Constraints, and Human-Centered Design

Beyond attendance monitoring, duty scheduling and student assistant assignment present complex challenges due to the need to satisfy multiple constraints, including student availability, class schedule conflicts, office requirements, and workload balance. Literature on algorithm-based scheduling demonstrates that fairness can be treated as an explicit objective while maintaining operational efficiency and improving workload distribution [11], [12].

However, algorithmic scheduling is not purely a technical concern. Studies emphasize its ethical and human-centered implications, particularly when scheduling decisions affect student well-being, time allocation, and academic balance [13]. As such, algorithm-driven systems should function as decision-support tools rather than fully automated decision-makers, ensuring that human judgment remains central to final assignments.

The design of SAMS follows this principle by producing a ranked list of recommended assignments while preserving administrative control over final decisions. In SAMS, the term “using algorithm” refers to a structured decision-support approach consisting of (1) rule-based filtering to ensure applicant eligibility and (2) weighted scoring and ranking based on skill matching, schedule compatibility, and workload balance. This transparent input–process–output structure promotes consistency and fairness while ensuring that assignment criteria remain explainable and defensible during evaluation [11], [12].

2.3.3 Software Quality Evaluation: ISO/IEC 25010 as an Assessment Framework

To ensure systematic and credible system evaluation, software quality assessment must be grounded in recognized standards. ISO/IEC 25010:2023 provides a comprehensive quality model covering functional suitability, performance efficiency, usability, reliability, and security—dimensions that are directly applicable to systems handling real-time monitoring, identity-linked attendance, and administrative decision support [14].

Anchoring system evaluation on ISO/IEC 25010 strengthens the validity of assessment outcomes by extending beyond subjective user satisfaction measures. This framework is particularly suitable for SAMS, as evaluation is conducted within a limited academic term, requiring measurable and defensible quality indicators aligned with international standards [14].

2.4 Technological Foundations Relevant to SAMS system

2.4.1 Cross-Platform Development and System Tooling

The Student Assistant Management System is implemented as a web- and mobile-based platform, necessitating technologies that support cross-platform consistency, scalability, and efficient development cycles. Flutter is adopted as the primary mobile development framework because it allows a unified codebase for both Android and iOS platforms, ensuring consistent user experience while reducing development time and maintenance effort [15].

Backend operations are supported by a combination of PHP and Node.js, which manage authentication, scheduling validation, algorithm-based recommendation processing, document generation, and API communication. This modular backend architecture supports scalability and long-term maintainability. The system's data layer utilizes MySQL to store structured records such as user profiles, attendance logs, duty schedules, application requirements, and generated documents, ensuring reliable and secure data management across platforms [15].

2.5 Supporting Libraries for Attendance, Authentication, and Document Outputs

To meet the functional requirements of the Student Assistant Management System (SAMS), the system integrates several third-party libraries that support attendance monitoring, secure authentication, and automated document generation. These libraries are used to enhance efficiency, accuracy, and security in system operations.

A QR scanning library is utilized to enable fast and contactless attendance logging, reducing manual input and minimizing recording errors during time-in and time-out transactions. To strengthen identity verification, an OTP generation and validation module is implemented as part of the system's multi-factor authentication mechanism, ensuring that attendance logs and sensitive transactions are linked to authenticated users only.

For document automation, a PDF generation toolkit is employed to produce standardized digital outputs such as Memoranda of Agreement (MOA) and Notices of Acceptance (NOA), supporting paperless administrative workflows and consistent document formatting. In addition, an authentication service is used to manage user sign-in, session handling, and identity validation across the mobile and web platforms.

These technology choices directly support the core objectives of SAMS, particularly secure time-in and time-out recording, automated document processing, and centralized administrative reporting, which collectively reduce manual workload and improve operational efficiency.

2.6 Research Gap

The reviewed literature collectively indicate that existing solutions remain “functionally fragmented.” Attendance solutions commonly lack scheduling, evaluation, and document management; scheduling solutions do not integrate secure attendance validation or monitoring; and document systems do not include duty assignment and performance tracking.

This gap is operationally significant for NU Lipa because student assistantship management requires coordination among multiple offices, time-sensitive attendance validation, and consolidated reporting for SDAO monitoring. SAMS addresses this gap by integrating secure attendance (QR + PIN/OTP), scheduling, document

2.7 RELATED SYSTEMS

This section presents a comprehensive review of existing systems and applications related to attendance monitoring, document management, scheduling, and system security, which are core functional areas relevant to the implementation of a student assistantship management platform. The discussion focuses on identifying the strengths and limitations of these systems in relation to the operational requirements of a university-based Student Assistant Program. Through this review, a gap analysis is conducted to determine how the proposed Mobile-Based Student Assistant Management System (SAMS) addresses the shortcomings observed in existing solutions, as emphasized in the reviewed literature and studies. The proposed system integrates attendance, scheduling, document workflows, and evaluation into a unified mobile and administrative web platform

Features	DepEd E-Docs (2022)	Smart Attendance System (2023)	MMSU Scheduler (2024)	GSJ QR Attendance (2024)	Mobile-Based Student Assistant System with Web Administrative Dashboard (Proposed)
Attendance Monitoring	☐	☒	☐	☒	☒
Scheduling	☐	☐	☒	☐	☒
Document Management	☒	☐	☐	☐	☒
Evaluation Module	☐	☐	☐	☐	☒
Mobile Integration	☐	☒	☐	☐	☒
Security (PIN / QR / OTP)	☐	☒	☐	☒	☒

Table 1: Gap Analysis Table of Related Systems and Applications

Discussion of Related Systems

1. DepEd E-Docs System (2022) The DepEd E-Docs System focuses on promoting paperless workflows by supporting electronic document processing and the use of digital signatures in official transactions. This system is effective in improving document authenticity, reducing manual paperwork, and strengthening transparency in administrative processes. However, it is not designed for operational workflows specific to student assistantship programs. In particular, it does not include attendance monitoring, duty scheduling, mobile-based student application handling, or performance evaluation modules. As a result, while DepEd E-Docs strengthens document workflow integrity, it cannot support the end-to-end management needs of a student assistant program where attendance validation, scheduling, and monitoring must be integrated with document handling.

2. Smart Attendance System (2023) Introduced QR code-based attendance logging and real-time notifications, improving accuracy and efficiency compared to traditional manual logging. It demonstrates how QR-enabled attendance reduces administrative workload and improves record reliability. Despite

these strengths, the system is primarily attendance-centered and lacks a centralized administrative dashboard and integrated modules for duty scheduling, document management, and student assistant evaluation. This limitation results in a fragmented solution because student assistantship management requires more than attendance—it also requires assignment validation against class schedules, duty-hour reporting, and monitoring tools linked to performance tracking.

3. MMSU Scheduler (2024) The Mariano Marcos State University (MMSU) Scheduler focuses on algorithm-driven scheduling to improve fairness and reduce conflicts in academic timetabling. This type of system demonstrates the value of automation in managing constraints and producing consistent scheduling outputs. However, the MMSU Scheduler is not designed to support secure attendance validation, mobile integration for student assistants, or document workflows required in a student employment program. Since the student assistantship context requires coordination between availability, attendance logs, office assignments, and administrative monitoring, a scheduling-only solution remains insufficient for end-to-end program management.

4. GSJ QR Attendance System (2024) The GSJ QR Attendance System highlights secure attendance logging through authentication and related security mechanisms. It supports the argument that QR-based attendance improves efficiency but should be strengthened with verification controls to protect data integrity and prevent proxy attendance. However, like many attendance-focused platforms, it lacks scheduling, evaluation, and administrative decision-support functionalities that are essential for student assistantship management. Without integrated scheduling and monitoring tools, attendance records remain isolated and cannot fully support duty assignment decisions, performance evaluation, and consolidated reporting requirement

2.8 SYNTHESIS

The reviewed literature and related systems indicate that although digital solutions have been increasingly adopted in educational institutions, most existing systems remain functionally fragmented. Attendance monitoring solutions primarily focus on logging efficiency but commonly lack integrated scheduling, document management, and performance evaluation capabilities, while scheduling systems often operate independently and do not incorporate secure attendance validation or real-time monitoring features.

Furthermore, the reviewed studies emphasize that efficiency alone is insufficient when attendance data is linked to accountability and evaluation. Literature highlights the necessity of strengthening QR-based attendance systems with additional authentication mechanisms, such as PIN or One-Time Password (OTP) validation, to prevent proxy attendance and unauthorized access. Related cybersecurity guidelines further support the adoption of multi-factor authentication (MFA) as a best practice for protecting systems that handle sensitive records, including attendance and identity-based transactions.

In addition, studies on algorithm-driven scheduling underscore the importance of fairness, workload balance, and human-centered decision support in environments where assignments affect individuals' time and responsibilities. These studies recommend algorithm-guided approaches that assist administrators through ranking or recommendation mechanisms while retaining human control over final decisions. To ensure that such systems are reliable and acceptable for institutional use, structured evaluation using recognized standards, such as ISO/IEC 25010, is recommended to assess usability, reliability, performance efficiency, and security of academic information systems.

Based on these insights, the proposed NU Lipa Student Assistant Management System (SAMS) directly addresses the identified gaps by integrating a mobile platform for students and a web-based administrative system, algorithm-driven duty assignment, secure attendance validation through QR and PIN/OTP mechanisms, digital document management, and performance evaluation tools within a unified platform. Consequently, the proposed system is not a duplication of existing work, but an innovative continuation that consolidates best practices from related literature and systems and introduces features specifically tailored to the operational needs of the Student Assistant Program at NU Lipa under the supervision of the Student Development and Activities Office (SDAO)

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technologies, components, and architecture used in the development of the proposed Student Assistant Management System. It includes detailed discussions on software, hardware, network, and peopleware requirements, as well as the system's technical design and special features.

4.1 DETAILS OF THE TECHNOLOGY TO BE USED

This section discusses the software, hardware, network, and peopleware requirements necessary for the development, testing, and deployment of SAMS. It covers the selected development tools, runtime environments, third-party libraries, deployment requirements, and project team roles required to execute the system successfully within one academic term.

Software/Hardware Requirements

SAMS is implemented as a mobile-based platform for student assistants and a web-based administrative dashboard for administrators and supervisors. The system requires tools that support cross-platform mobile development, web dashboard development for program management, API services, secure authentication, QR-based attendance scanning, OTP validation, and automated document generation (PDF outputs). The selected software stack is designed to balance feasibility, scalability, maintainability, and ease of deployment within an academic environment.

Table 2. Software Requirements for Systems Development

Software Classification	Software Name	Description / Purpose
Integrated Development Environment (IDE)	Visual Studio Code, Android Studio	Primary environments for coding, debugging, building mobile app (Flutter) and web/admin modules.

Programming Languages / Frameworks	Flutter, PHP, Node.js, HTML, CSS, Java	Flutter for cross-platform mobile app; PHP/Node.js for backend logic and APIs; web technologies for admin dashboard UI.
Database	MySQL	Relational storage for users, schedules, attendance logs, evaluation results, and generated documents.
Third-Party Libraries	QR Code Scanner, OTP Generator, PDFKit, Firebase Auth	QR scanning for attendance; OTP generation/validation; PDF outputs for MOA/NOA; authentication support.

Version Control / Collaboration	GitHub	Source control, collaboration, and release management during development.
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Table 3. Hardware Requirements for Systems Development

Hardware Component	Technical Specifications	Description / Purpose
Development Laptop / Workstation	Intel Core i5, 8GB RAM, 256GB SSD	Primary device used for coding, testing, and building mobile/web modules.
Smartphone Devices	Android 8+, iOS 12+	Used to test QR scanning, OTP validation, and mobile attendance features.
Server Hosting	Cloud-based VPS, 2 vCPU, 4GB RAM, 50GB SSD	Hosts backend services/APIs and the MySQL database for production or pilot deployment.
QR Scanner Module	Built-in camera (mobile device)	Captures QR codes for attendance logging.

Network Requirements

A stable Internet connection is required for system operations involving real-time synchronization between the mobile application and the web dashboard. Communication is performed through RESTful APIs secured using HTTPS to ensure confidentiality and integrity of data transmissions between users (student assistants, office heads, SDAO admins) and the backend services.

Minimum network needs include:

- Reliable Wi-Fi or mobile data connection for attendance submission, schedule retrieval, and document generation requests.
- Secure HTTPS API communication to reduce risks of data interception and tampering.

Peopleware Requirements

The development of SAMS requires a collaborative team with defined responsibilities across UI/UX design, backend development, documentation, and stakeholder coordination

Table 4. Project Team Members, Roles and Contact Details

Name(s)	Role	Contact Details
Niel Malleon	UI/ UX Designer & Documentation Lead	nmalleon1014@gmail.com
John Clarenz Dimazana	UI/UX Designer	dimazanajc10@gmail.com
Jaime III Garcia	Backend Developer	jgarvia9@gmail.com
Luke Aron Magpantay	UI/UX Designer	magpantaylukearon@gmail.com
Martyn Joseph Seloterio	Backend Developer	seloteriomartyn@students.nu-lipa.edu.ph
Name(s)	Role	Contact Details
Zaira Joy S. Enayo	SDAO Head	zsenayo@nu-lipa.edu.ph
Joseph Michael E. Aramil	Program Chair	jearamil@nu-lipa.edu.ph

4.2 PROJECT TECHNICAL DESCRIPTION

SAMS is a **mobile and web-based platform** designed to streamline student assistant operations through integrated modules for application processing, secure attendance monitoring, scheduling, document management, monitoring/reporting, and performance evaluation. The system's title includes "using algorithm" because it implements a rule-based filtering mechanism and a weighted scoring/ranking approach that assists administrators in consistent and fair assignment decisions.

Special Features of the System

The system introduces dual-layer security for attendance logging by combining QR code validation with PIN/OTP authentication, minimizing the risk of proxy attendance and unauthorized access. The algorithm-guided recommendation engine standardizes duty assignment by ranking applicants based on skills, schedule compatibility, and workload balance, reducing bias and improving

administrative efficiency. Additionally, automated document generation with PDF output and e-signature readiness streamlines administrative processing and supports paperless transactions. Real-time synchronization between the Flutter mobile application and web dashboard ensures data consistency across platforms.

The system incorporates multiple security mechanisms to address common risks associated with digital attendance logging and data management. QR codes used for attendance validation are designed to be session-based and time-limited, reducing the risk of reuse, screenshot sharing, or unauthorized access. Each QR scanning process is paired with PIN or OTP validation, ensuring that only authenticated users can successfully record attendance.

To mitigate risks such as phone borrowing and proxy attendance, OTP codes are configured with short expiration periods and are validated against active user sessions. All data transmissions between the Flutter mobile application, web-based admin dashboard, and backend services are secured using HTTPS protocols to protect data confidentiality and integrity. While advanced institutional-level data privacy integration is beyond the scope of this study, the implemented security measures demonstrate compliance with fundamental authentication and data protection principles suitable for academic information systems

System Architecture

The proposed system adopts a four-layer architecture to ensure clear separation of responsibilities and organized interaction among system components. The User Layer is composed of Student Assistants, Office Heads, and SDAO Administrators, who access the system according to their assigned roles and permissions.

These users interact through the Presentation Layer, which consists of a Flutter-based mobile application for student assistants and a web-based administrative dashboard for SDAO administrators and office heads. The mobile application supports student-facing transactions such as application submission, duty viewing, and attendance logging, while the web dashboard supports administrative monitoring, scheduling, and program oversight.

All requests from both interfaces are routed through an HTTPS REST API, which serves as the system's secure communication gateway and connects the presentation components to the backend services. Behind the API is the Logic Layer, where the system's core business services are executed, including requirement filtering, algorithm-guided recommendation for office assignment, scheduling

and conflict checking, QR plus PIN/OTP attendance validation, performance evaluation, biocalendar/workload visualization, and automated document processing for MOA/NOA with PDF output. This layer functions as the processing and decision engine, ensuring that the program rules and constraints are consistently applied across both mobile and web platforms. At the foundation is the Data Layer, which includes a MySQL database for storing structured records such as user credentials, attendance logs, duty schedules, and evaluation results, and a dedicated file storage component for uploaded requirements and generated PDF documents (e.g., MOA/NOA). This separation supports efficient handling of both transactional data and document files while improving maintainability and scalability.

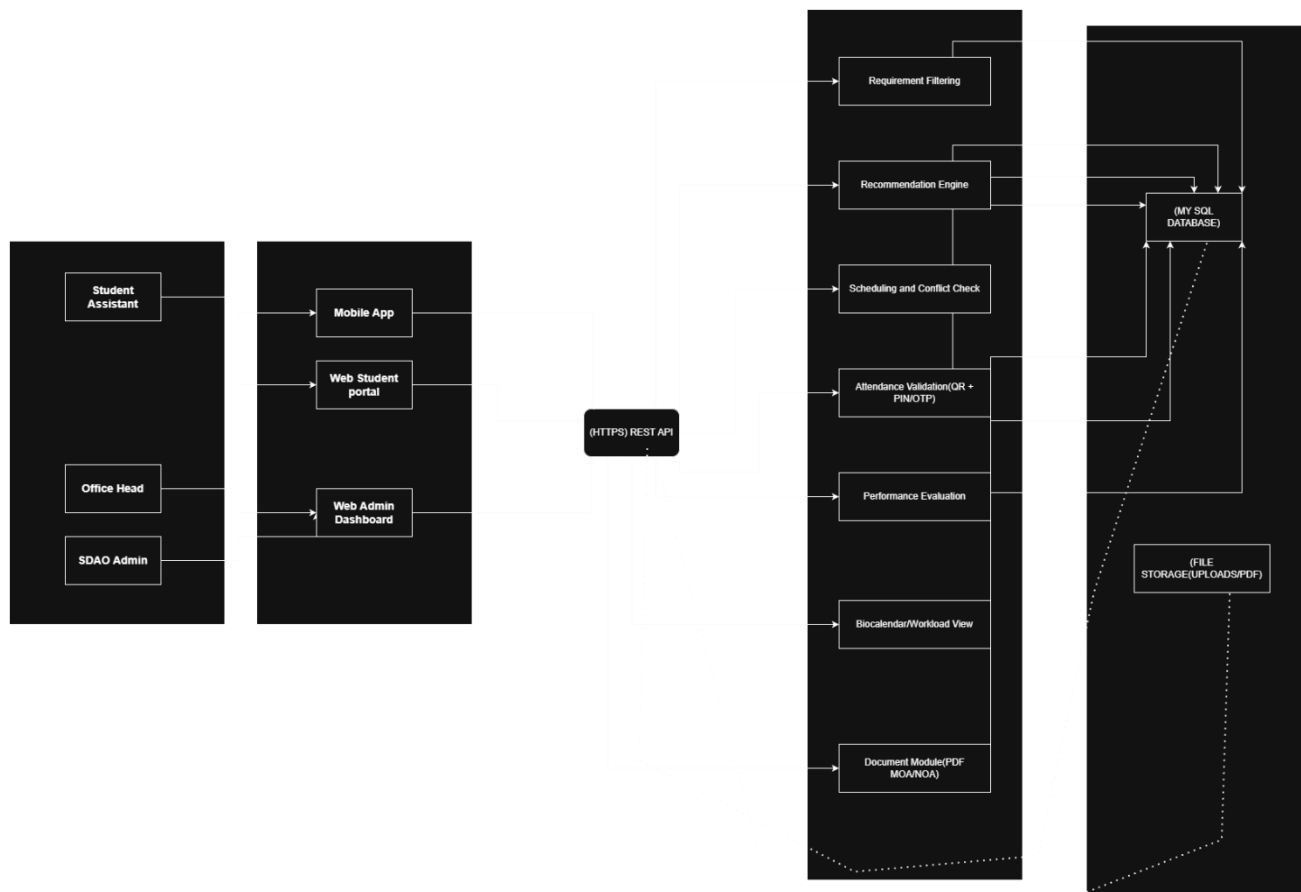


Figure 1. Overall System Architecture

Mobile Application Modules (MAA)

The mobile application modules are designed primarily for student assistants, providing a convenient platform for application submission, duty monitoring, and secure attendance logging. These modules run on Android and iOS devices using a Figma.

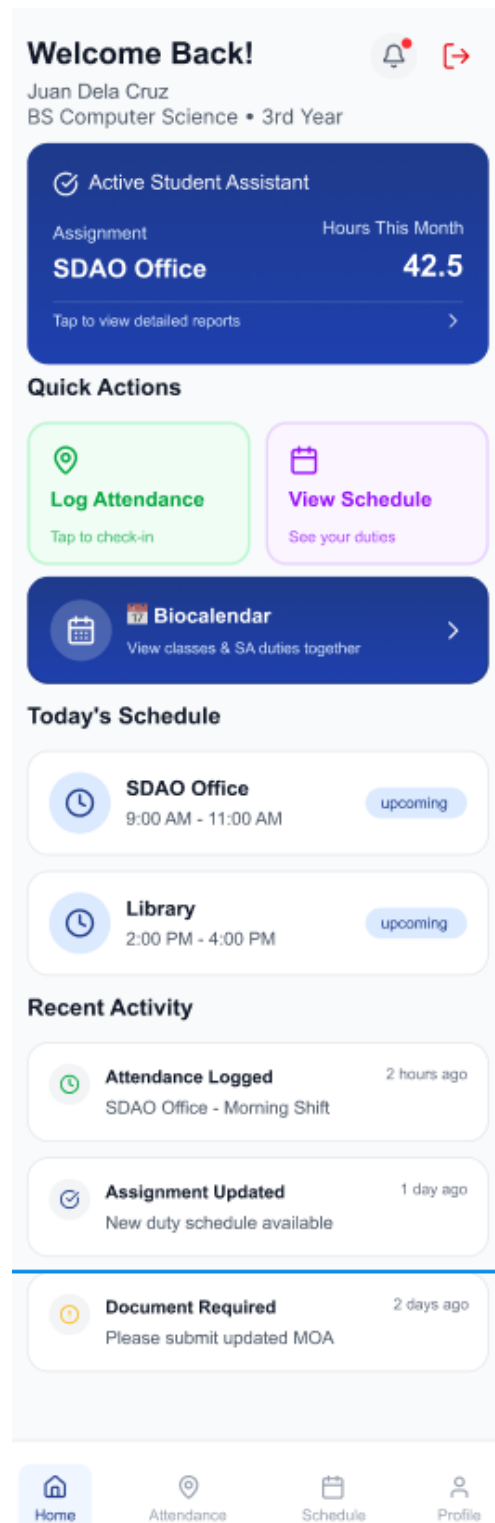
1. Authentication & Access Module:

The image displays three mobile application screens for the 'NU Lipa SA System'.

- Student Login Screen:** Features a blue header with a graduation cap icon and the text 'NU Lipa SA System Student Assistant Management'. Below, under the 'Student' heading, are input fields for 'Email Address' (containing 'student@nu-lipa.edu.ph') and 'Password' (with a placeholder 'Enter your password'). A blue 'Sign In' button is at the bottom, along with a 'Demo Mode: Click sign in to proceed' link.
- Create Account Screen:** Has a blue header with a back arrow and 'Create Account' title. It includes a subtitle 'Join NU Lipa SA System' and several input fields: 'First Name' (Juan), 'Last Name' (Dela Cruz), 'Student ID' (2021-00001), 'Email Address' (your.email@example.com), 'Password' and 'Confirm' (both masked with asterisks), and 'Contact Number' (+63 912 345 6789). There are also dropdowns for 'Program' and 'Year Level', and a 'Skills' section with checkboxes for Microsoft Office, Event Management, Communication, Time Management, Leadership, Social Media, and Design. A blue 'Continue to Verification' button is at the bottom, followed by a small disclaimer: 'By registering, you agree to the terms and conditions'.
- Verify Email Screen:** Features a blue header with a back arrow and a shield icon. The title is 'Verify Your Email', with a subtitle 'We've sent a 6-digit code to example@email.com'. Below is a 6-digit OTP input field. A blue 'Verify Email' button is present, along with a 'Resend OTP Code' link. At the bottom, it says 'Demo: Use code 123456 to verify'.

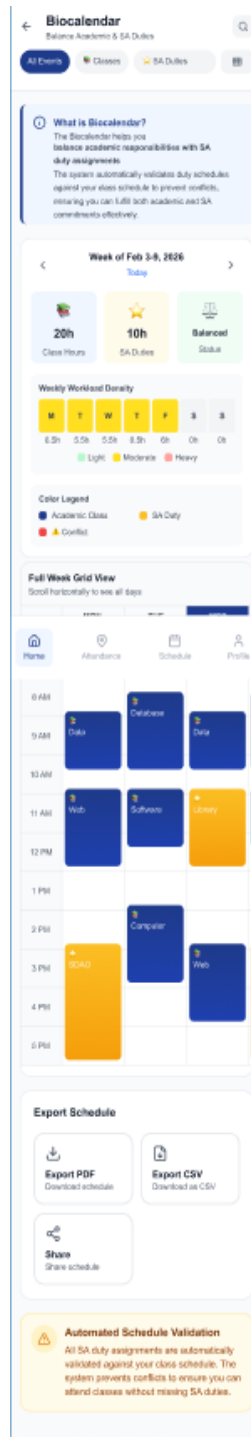
- Manages secure access to the mobile application for registered student assistants.
- Implements account authentication combined with PIN or One-Time Password (OTP) validation to enhance security.
- Ensures that only authorized users can access attendance, scheduling, and application features.

2. Student Dashboard Module:



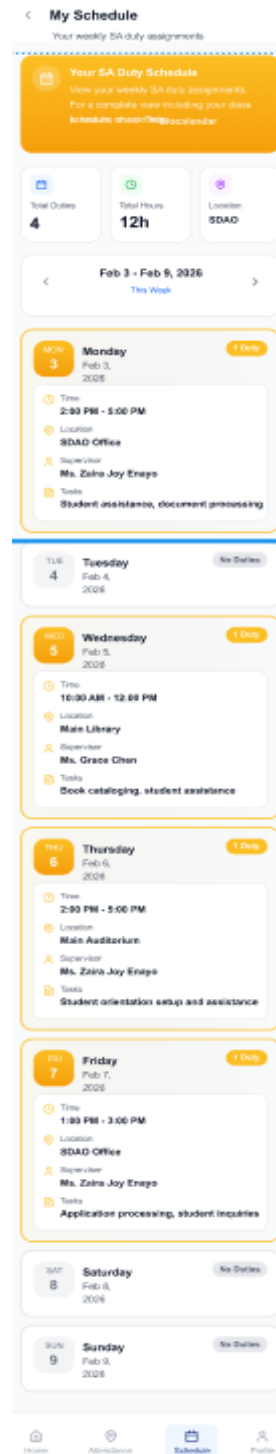
- Serves as the main landing interface after successful authentication.
- Displays assigned office, and summary of duties and attendance.
- Provides centralized navigation to attendance logging and schedules

3. Duty Schedule Viewing Module:



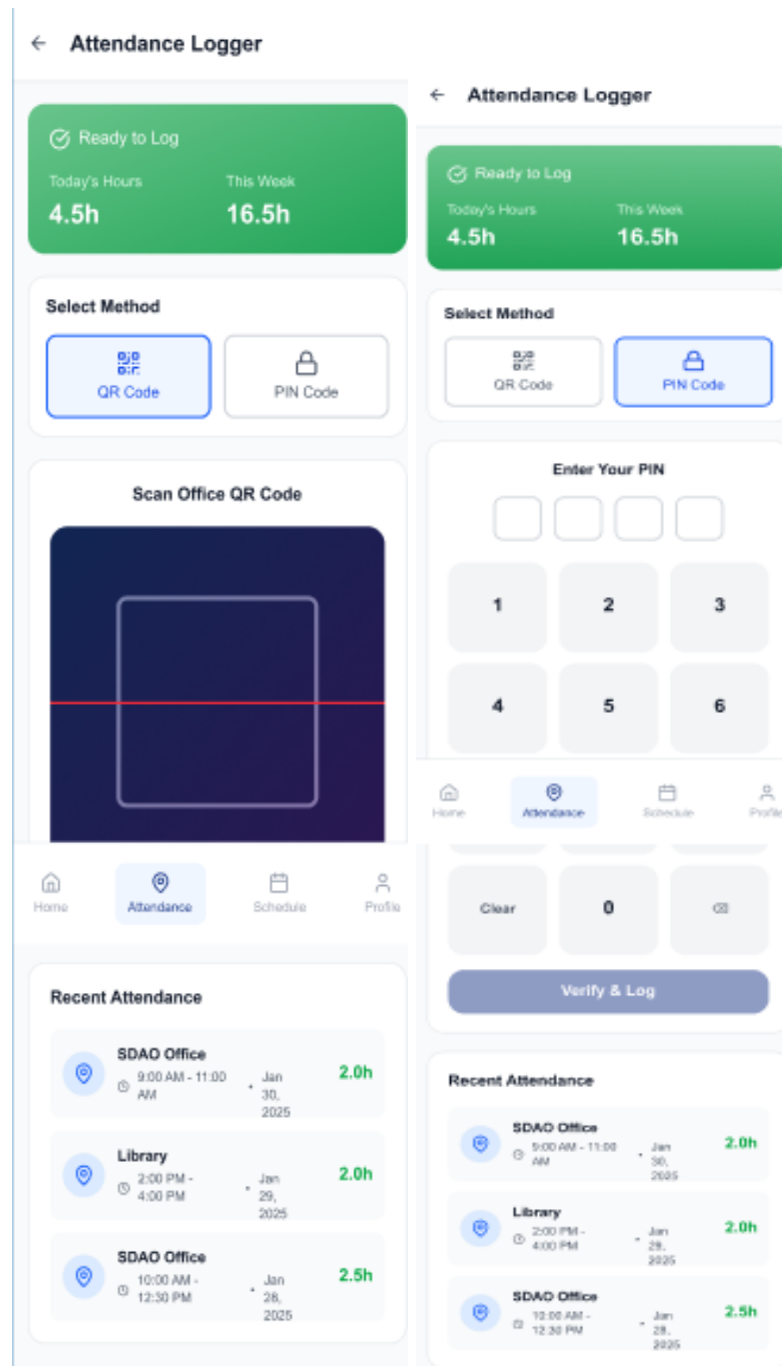
- Displays assigned duty schedules validated against the student's submitted class schedule.
- Presents daily and weekly duty assignments, including office location and time slots.

4. Duty-Hour Monitoring Module:



- Displays total rendered duty hours

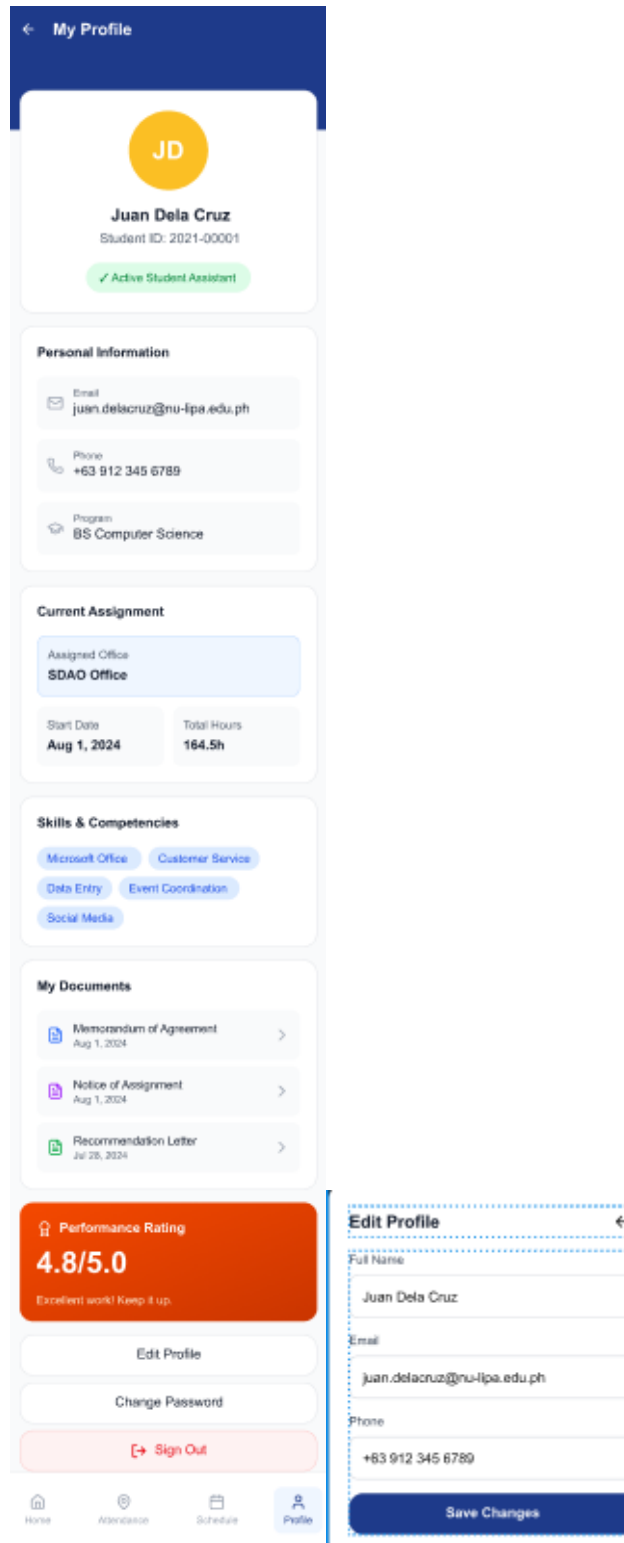
5. Attendance Logging Module:



- Implements secure digital time-in and time-out functionality.
- Utilizes QR code scanning combined with PIN to prevent proxy attendance.

- Submits attendance records to the backend system via a secure HTTPS REST API.

6. User Profile Management Module:



- Profile Screen

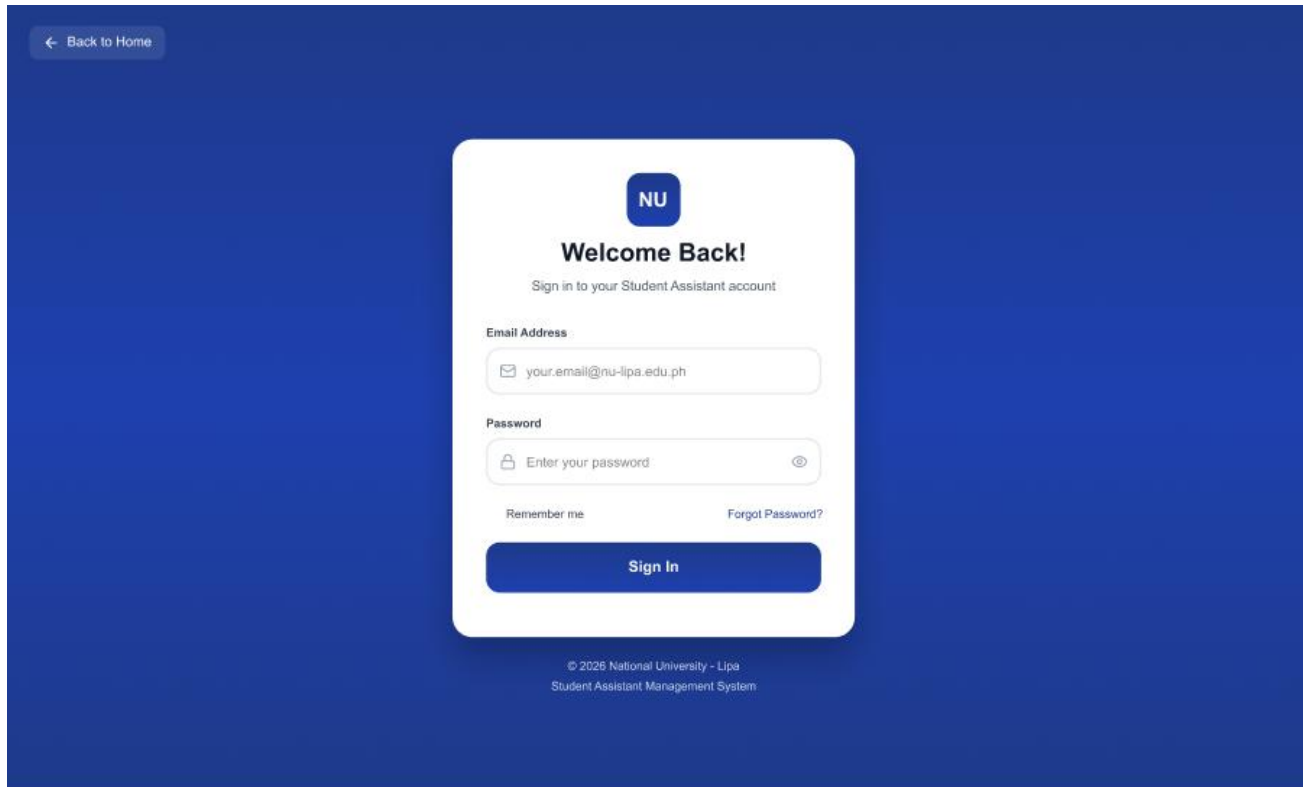
- Edit Profile Screen

Web Application Modules (MWA)

Administrative Dashboard

These modules are accessible through a web browser and are designed for administrators and office heads responsible for managing the Student Assistant Program. The admin dashboard supports application evaluation, duty scheduling, attendance monitoring, document management, and performance evaluation. It communicates with backend services through a secure HTTPS REST API to ensure real-time monitoring, data integrity, and controlled administrative access.

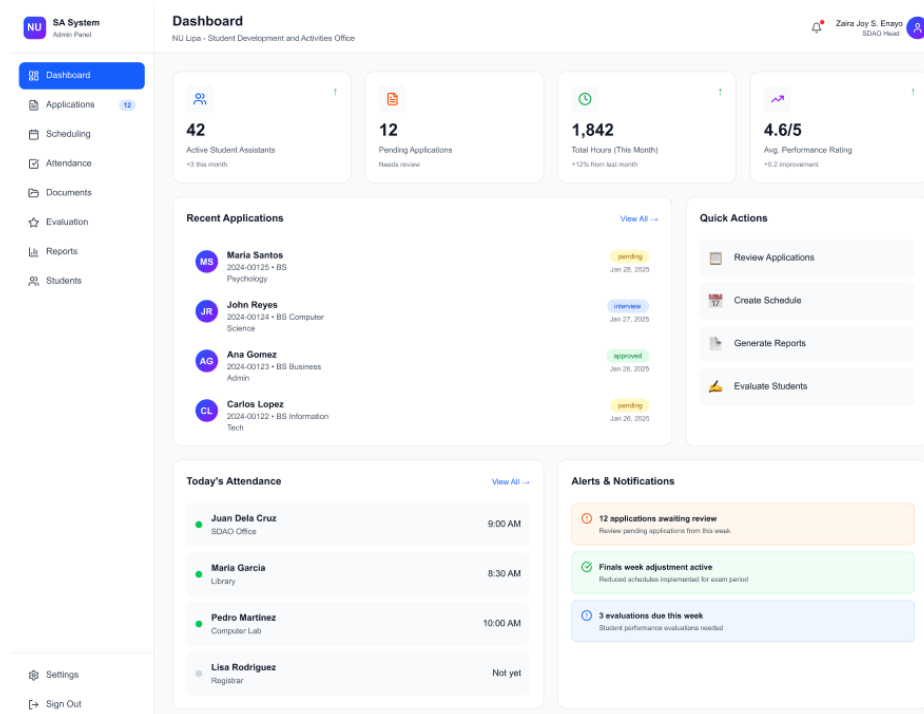
1. Admin Authentication & Access Control Module:



The screenshot displays a login interface for the Student Assistant Management System. It features a dark blue background with a central white login card. At the top left of the card is a 'Back to Home' button. The card includes the 'NU' logo, a 'Welcome Back!' greeting, and a prompt to 'Sign in to your Student Assistant account'. Below this are input fields for 'Email Address' (containing 'your.email@nu-lipa.edu.ph') and 'Password' (with a placeholder 'Enter your password' and a toggle icon). There are also links for 'Remember me' and 'Forgot Password?'. A prominent blue 'Sign In' button is at the bottom of the card. The footer of the page states '© 2026 National University - Lipa Student Assistant Management System'.

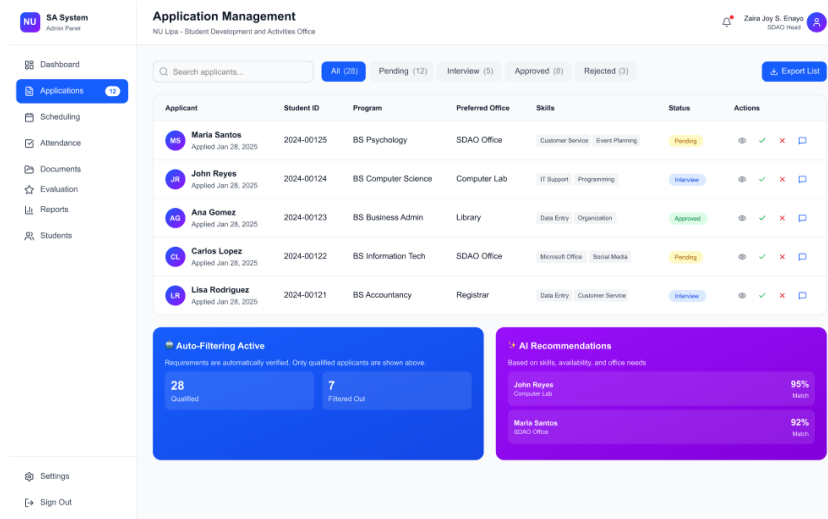
- Allows SDAO administrators and office heads to log in.
- Ensures that only authorized personnel can access administrative features.
- Serves as the primary entry point of the admin dashboard.

2. Administrative Dashboard & Monitoring Module:



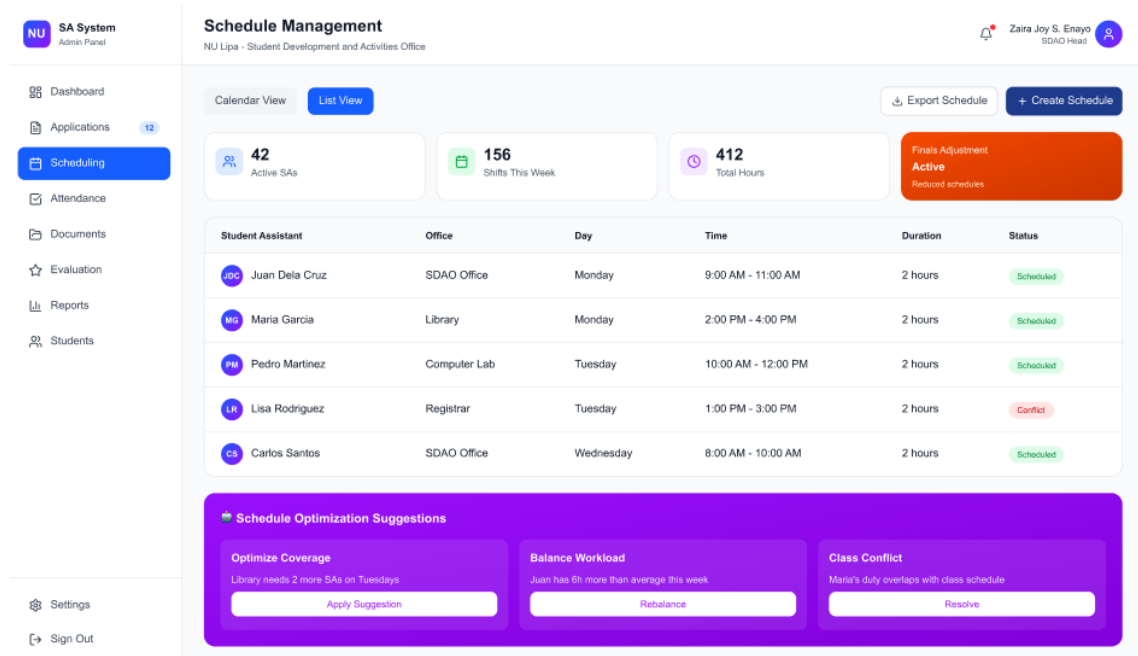
- Shows statistics such as number of active student assistants and pending applications..
- Provides administrators with real-time visibility of program status.

3. Student Application Module:



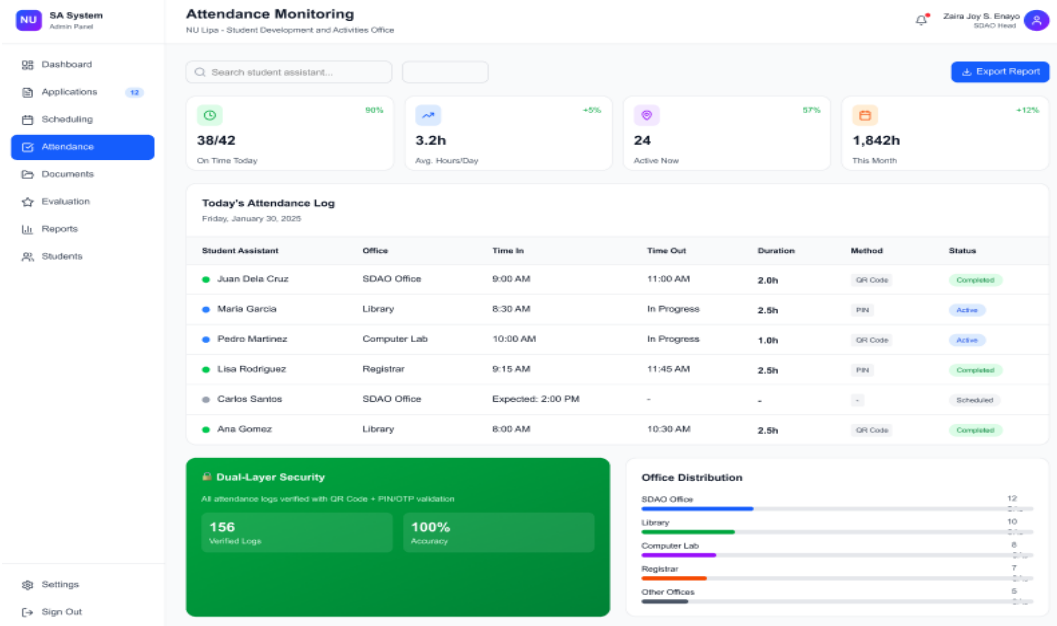
- Displays a list of student assistant applicants.
- Supports filtering by status, qualifications, and submitted requirements.
- Enables efficient review of large volumes of applications.

4. Duty Scheduling & Assignment Management Module



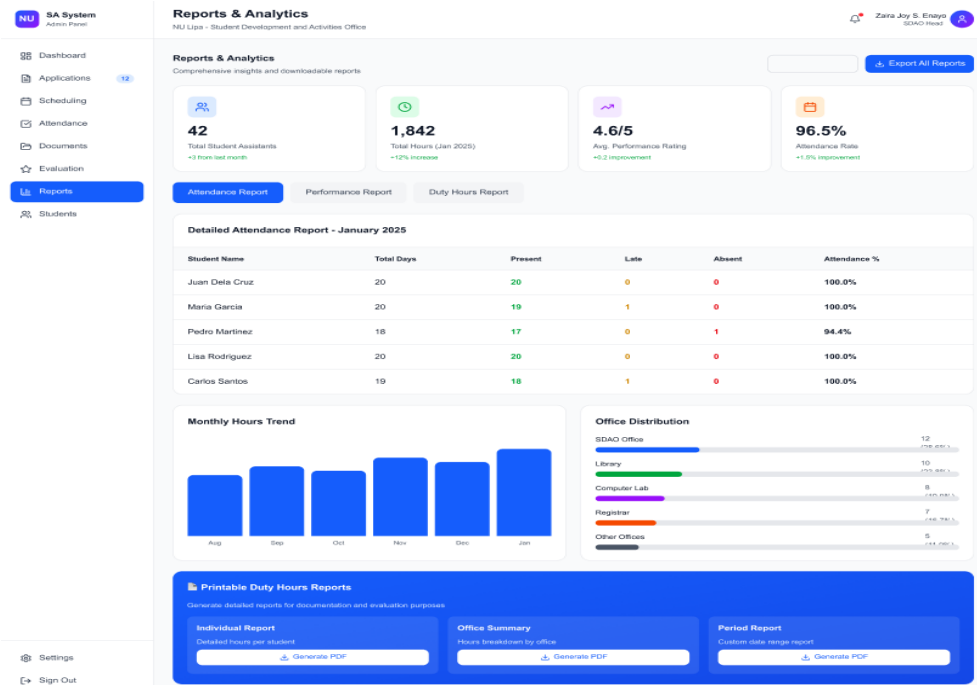
- Allows administrators to assign student assistants to offices and time slots.
- Validates schedules against student class availability.
- Prevents scheduling conflicts and workload imbalance.

5, Attendance Monitoring



- Displays live attendance data from student assistants.
- Shows time-in and time-out records as they are logged.

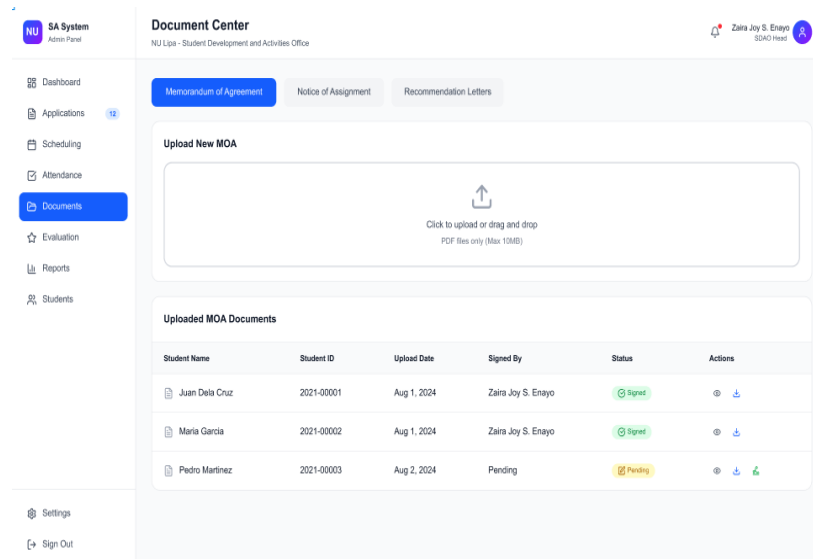
6. Reporting & Duty-Hour Management Module:



- Displays computed duty-hour totals per student assistant.

- Supports generation of consolidated attendance and duty reports.

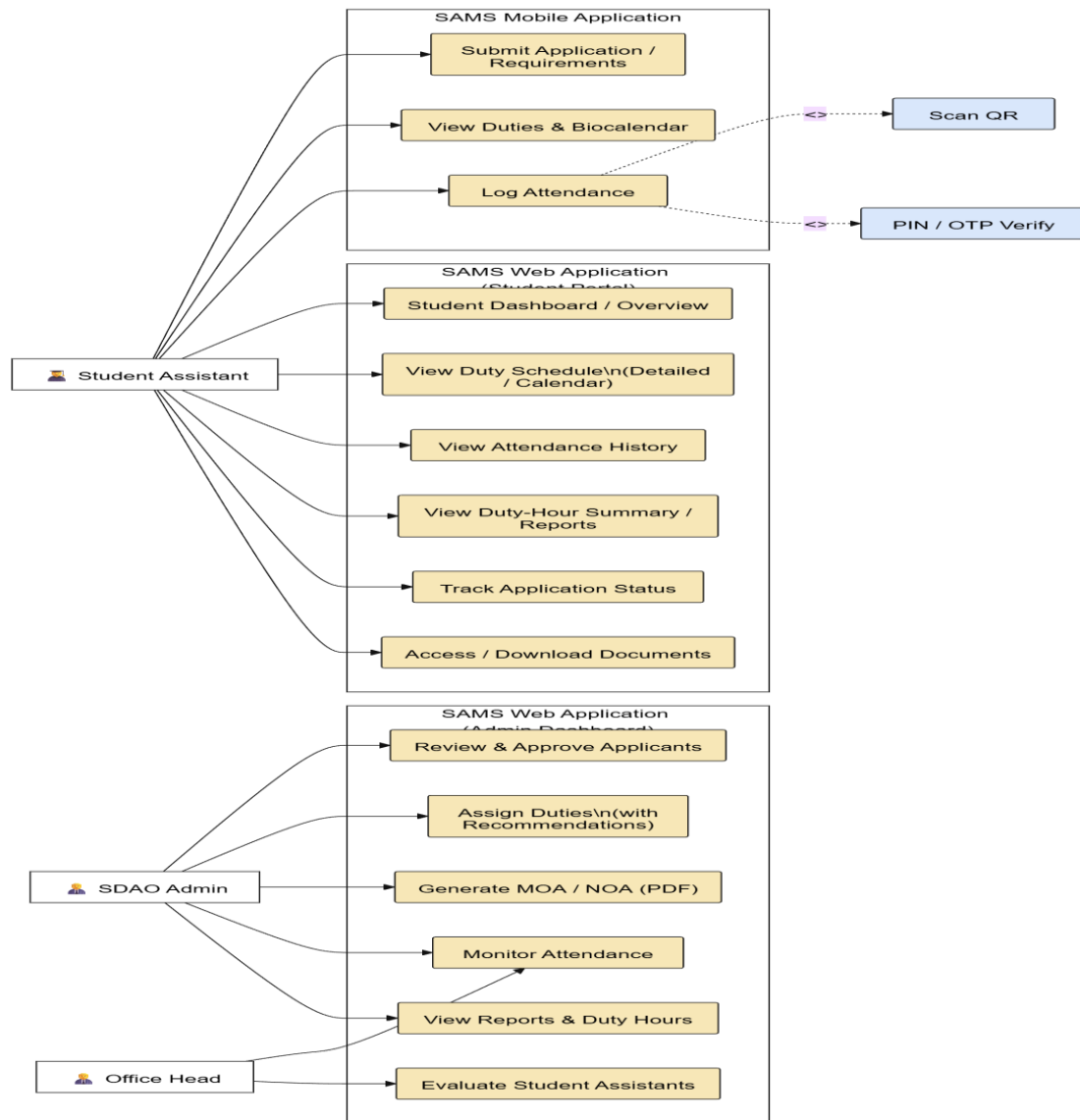
7. Document Management:



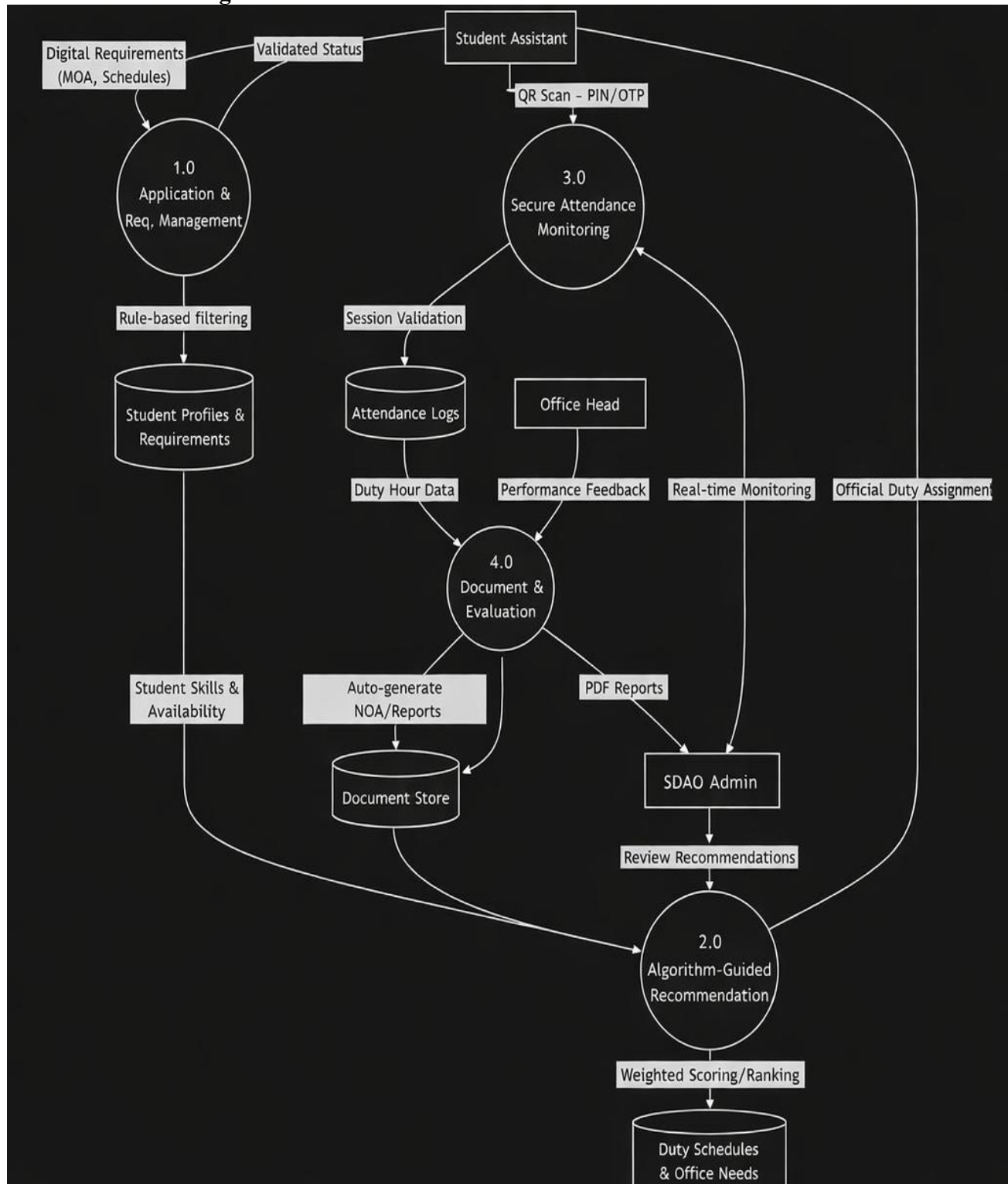
- Manages uploaded student requirements and supporting documents.
- Displays verification status and document history.

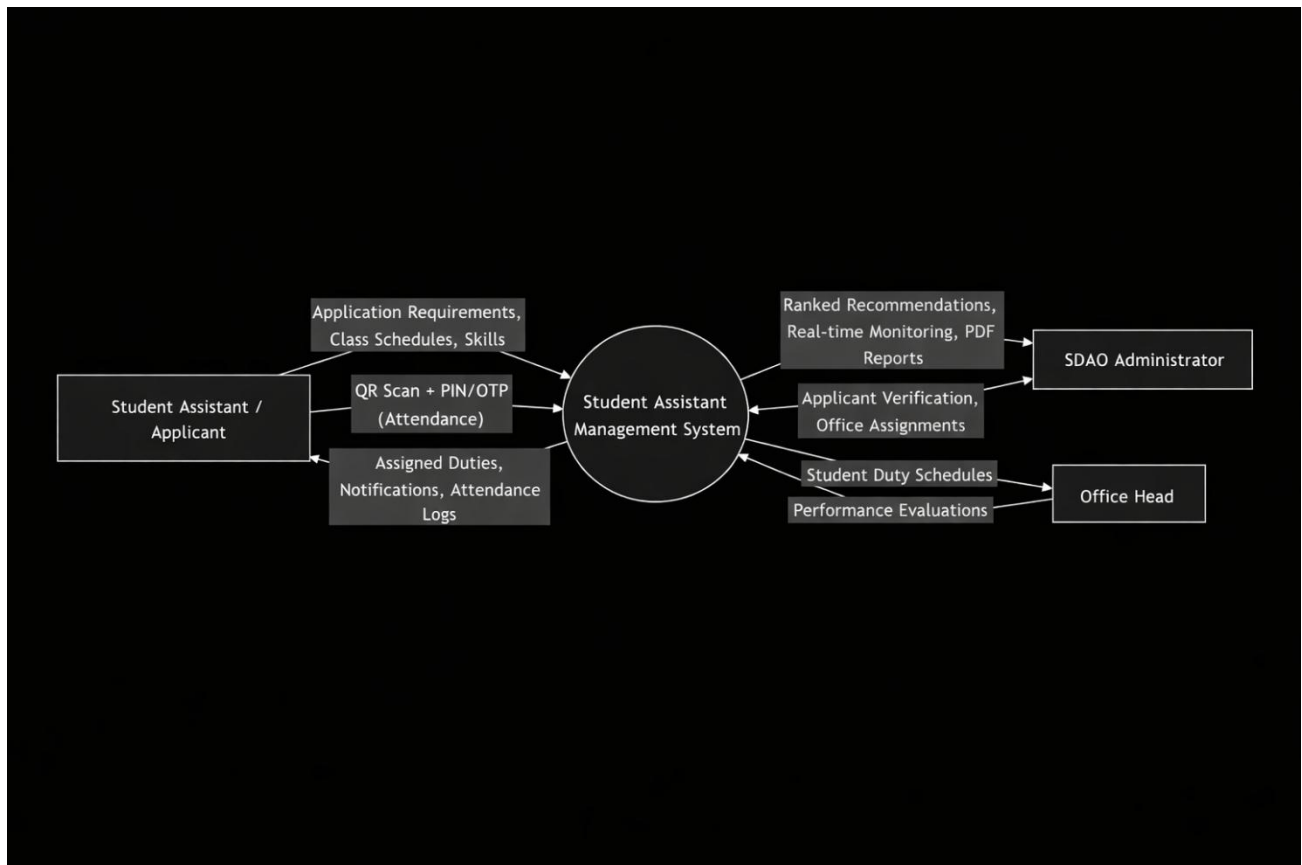
Supporting Diagrams:

SD.1 UML (Unified Modeling Language) Diagram of SAMS



SD.2 Data-Flow-Diagram of SAMS

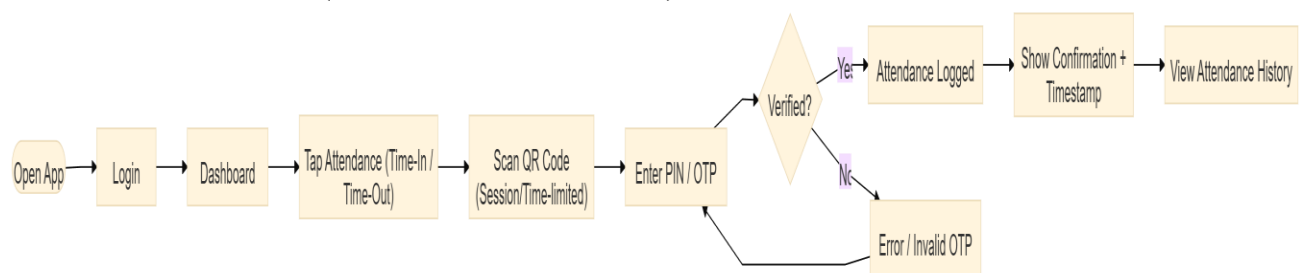




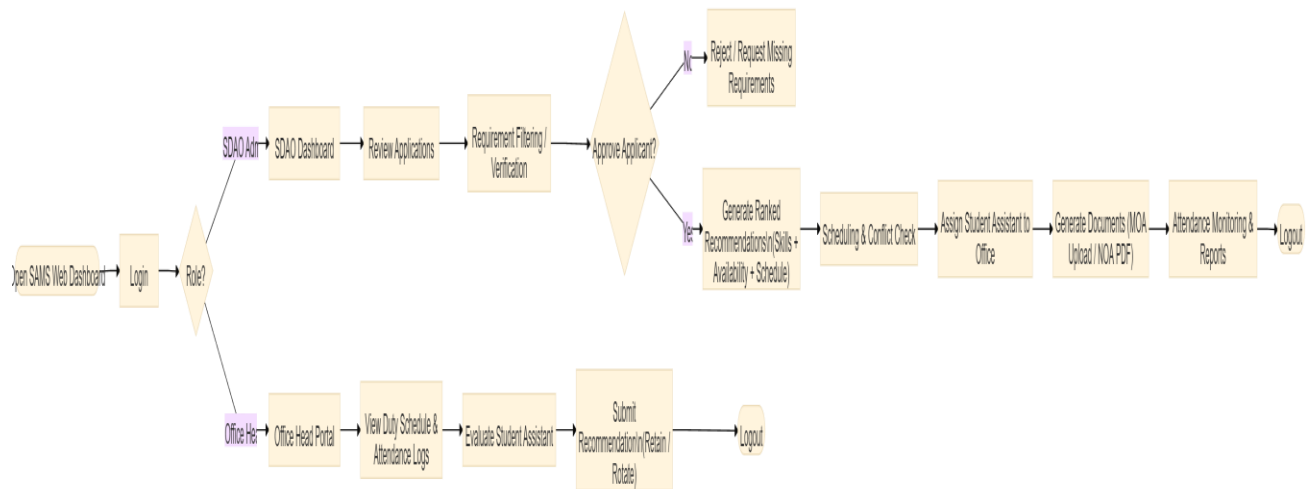
SD.3 Mobile User Flow (Student Assistant)



SD.4 Mobile User Flow (Attendance – Own Device)



SD.6 Web User Flow (SDAO Admin + Office Head)



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